

Team name:	UNIQA CRO TEAM
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#	Name	Software's Tool/Library	Description
1	Python 4	Programming Language	Main language for the entire robot controller (~
2	Webots	Simulation Platform	3D robot simulator used as the base for the Er
3	Erebus	Simulation Platform	RoboCupJunior Rescue Simulation environme
4	NumPy	Library	N-dimensional array operations, 5x5 cell matri
5	OpenCV (cv2)	Library	Camera frame processing, BGR-to-HSV conver
6	PyTorch	Library	CUDA device detection and GPU memory ma
7	Ultralytics YOLO (YOLOv8)	Library / AI Framework	YOLOv8 model training pipeline and real-time
8	best_erebus.pt (Custom YOLO	AI Model	Custom-trained YOLOv8 model (7 classes: C,
9	Custom Victim Dataset	AI Dataset	Labeled image dataset collected from Erebus
10	Custom Robot Controller (main	Software	Full autonomous navigation, mapping, and det
11	struct (Python stdlib)	Library	Binary message packing for victim reports (int
12	colorsys (Python stdlib)	Library	HSV colour conversion for colour sensor class
13	math (Python stdlib)	Library	Trigonometry for turn angle calculations (atan2
14	copy (Python stdlib)	Library	Deep-copying to _explore sets and path sets to
15	Visual Studio Code	Software Tool	Primary code editor used for development and
16	Git / GitHub	Software Tool	Version control and collaborative development
17	LabelImg	Software Tool	Image annotation tool used to label victim and
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